Conditional Diffusion for Medical Image Dehazing with GAN-Prior

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Abstract. Deep learning-based medical image dehazing often requires paired clean-noisy data for supervised training, which is notoriously difficult to acquire in clinical practice. Existing unpaired methods, while avoiding this requirement, often suffer from high model complexity and insufficient detail recovery. Method: To address these challenges, we propose a novel unpaired dehazing framework named NIL-Diffusion. Our approach consists of two cooperative stages: (1) A Noise-Imitation Learning (NIL) module, which employs a streamlined adversarial network (only one generator and one discriminator) to extract noise features and generate an initial denoised estimate. (2) A Conditional Diffusion Refinement module is proposed to perform precise, detail-aware restoration through a reverse diffusion process. In this module, the initial denoised result from the generator serves as the input to the forward diffusion process, where noise is progressively added. The original noisy image is then utilized as a conditioning input to guide the reverse diffusion process, enabling iterative refinement of the preliminary result towards a higher-quality restoration. Furthermore, we introduce a perceptual loss by leveraging a pre-trained Ultrasound Foundation Model (USFM) as a feature extractor, calculating the L1 distance between features of the clean and final denoised images to enforce semantic consistency.

Keywords: Medical Image Dehazing, Unpaired image, Conditional Diffusion

1 Architecture

1.1 NIL Framework

The Noise-Imitation Learning (NIL) framework [1] builds upon two well-established architectures: the NAFNet for the generator and a PatchGAN for the discriminator. The generator, based on NAFNet, is responsible for the core tasks of noise extraction and initial image reconstruction. For the adversarial discriminator, we employ the PatchGAN structure. To enhance the stability of the adversarial training process and mitigate common issues such as mode collapse, we incorporate Spectral Normalization (SN) into all layers of the PatchGAN discriminator. This modification effectively constrains the Lipschitz constant of the discriminator, leading to more reliable gradient feedback for the generator during training.

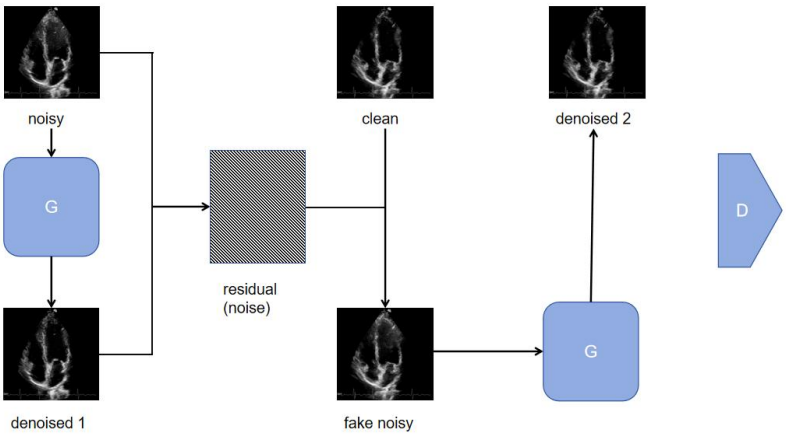

Fig. 1. NIL framework. First, the real noisy image undergoes preliminary denoising to obtain denoised1. The estimated residual (noise) is then obtained by subtracting denoised1 from the real noisy image. This estimated residual is added to unpaired real clean images to generate fake noisy images. The fake noisy images are subsequently denoised through the generator to produce denoised2. Finally, the clean images, denoised1, and denoised2 are collectively fed into the discriminator.

Furthermore, to ensure semantic consistency and enhance the perceptual quality of the denoised output, we introduce an additional perceptual loss term based on the Ultrasound Foundation Model (USFM) [2]. Specifically, the pre-trained USFM serves as a feature extractor, where both the ground truth clean images and the final denoised images are fed through this network. The L1 distance between the corresponding feature representations is then computed and minimized, effectively guiding the generator to produce results that are not only structurally accurate but also semantically meaningful in the context of medical imaging. This integration of a domain-specific foundation model provides strong perceptual constraints that complement the adversarial training process, particularly important for preserving diagnostically relevant features in medical image dehazing tasks.

The NAFNet (Non-Activation Feed-forward Network) represents a recent and significant advancement in neural network architecture for image restoration tasks. Departing from traditional designs that rely on complex activation functions and attention mechanisms, NAFNet proposes a simplified yet highly effective baseline. Its core innovation lies in the introduction of a Non-Activation Block, which removes non-linear activation functions altogether. Furthermore, it simplifies the channel attention mechanism into a computationally efficient, scalar-based variant. This streamlined design results in a remarkable reduction in computational complexity and memory consumption while simultaneously achieving state-of-the-art performance on

various benchmarks. We adopt NAFNet as our generator due to its superior efficiency and performance, making it ideally suited for the initial stage of noise extraction and preliminary reconstruction in our NIL pipeline.

The PatchGAN discriminator serves as a cornerstone architecture in many image-to-image translation tasks facilitated by Generative Adversarial Networks. Unlike a standard discriminator that outputs a single scalar value representing the probability of an entire input image being real, the PatchGAN is fully convolutional and outputs a two-dimensional matrix of probabilities. Each element in this matrix corresponds to a specific receptive field in the input image, representing the discriminator's confidence regarding the authenticity of that local region. This design enables the discriminator to focus on high-frequency details and local texture patterns, effectively modeling fine-grained structures—a critical capability for assessing the realism of deblurred or dehazed images. By emphasizing local fidelity, the PatchGAN guides the generator to produce outputs that are not only globally coherent but also locally realistic.

Spectral Normalization (SN) is implemented as a powerful regularization technique to stabilize the training of GANs. Its primary objective is to constrain the Lipschitz constant of the discriminator function, which essentially measures how sensitive a function is to changes in its input. A constrained Lipschitz constant prevents the discriminator from becoming too powerful too quickly, thereby avoiding vanishing gradients and training divergence. Technically, SN achieves this by normalizing the weight matrix in each layer of the discriminator by its largest singular value, efficiently implemented using a fast power iteration method during each training step. The integration of SN into our PatchGAN discriminator offers enhanced training stability by mitigating common failure modes like mode collapse and oscillatory behavior, while also improving model performance through more meaningful gradient feedback to the generator, ultimately facilitating the generation of higher-quality outputs.

1.2 Conditional Diffusion Model

Denoising Diffusion Probabilistic Models, often simply called diffusion models, have emerged as a powerful framework for generative modeling in computer vision. Unlike generative adversarial networks that employ a single-step generation process, diffusion models operate through a progressive refinement mechanism. These models begin with pure random noise and gradually transform it into coherent images through a series of iterative denoising steps, typically achieving superior sample quality and diversity compared to previous approaches.

The conditional extension of diffusion models incorporates additional guidance information during this denoising process. By integrating conditional inputs such as text descriptions, semantic maps, or reference images, these models can exercise precise control over the generation process. This capability makes them particularly valuable for medical imaging applications where anatomical fidelity and spatial precision are critical.

Recent advancements have explored various strategies for implementing conditioning in diffusion models. Some methods employ feature-wise modulation techniques where conditional information is injected into the network's intermediate layers. Others utilize cross-attention mechanisms that allow the model to dynamically focus on relevant aspects of the conditioning input. These technical innovations enable the model to maintain strong spatial correspondence with the guidance information while performing the denoising operation.

1.3 NIL-Diffusion

The NIL-Diffusion framework introduces a two-stage approach for image denoising that synergistically combines the strengths of generative adversarial networks and conditional diffusion models. This method first employs a GAN-based module to obtain preliminary denoised images, which provide crucial prior knowledge about the denoising task. Subsequently, a conditional diffusion model utilizes this prior knowledge while being guided by the original noisy input image throughout the reverse diffusion process.

During the training phase, the framework implements a structured training strategy where the Noise-Imitation Learning (NIL) component and the conditional diffusion model are optimized in distinct stages. This phased approach ensures stable convergence and effective knowledge transfer between the two components. The NIL module first learns to generate initial denoising results, establishing a robust foundation for the subsequent diffusion process. The conditional diffusion model then learns to refine these initial results using the original noisy images as conditional guidance.

For inference, the framework employs an efficient single-model architecture that incorporates the knowledge acquired from both training stages. The complete system operates by taking a noisy input image and processing it through the optimized conditional diffusion model, which leverages both the learned prior knowledge from the GAN component and the conditional guidance from the input itself. This design ensures that the reverse diffusion process is effectively guided toward producing high-quality denoised outputs while maintaining computational efficiency during deployment.

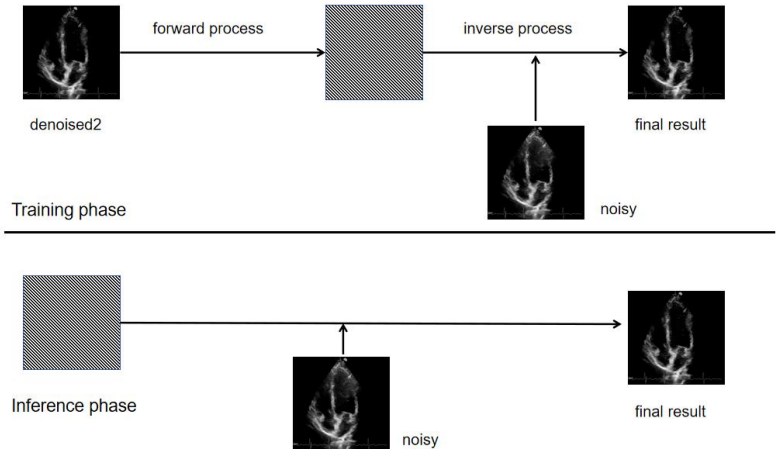

Fig. 2. NIL-Diffusion

2 Experiments and Results

2.1 Data

This challenge is established to advance the research in echocardiography image dehazing, particularly focusing on patients who present typical difficulties in clinical imaging. The dataset comprises a total of 6,900 four-chamber echocardiography images, including 4,500 images from 75 easy-to-image patients and 2,400 images from 40 difficult-to-image patients. All data were acquired under approved protocols and anonymized to ensure privacy.

For experimental purposes, the dataset is partitioned in a structured manner to facilitate both training and validation. The clean images are divided into training and validation sets at an 8:2 ratio. Noisy images are partitioned based on the availability of corresponding masks: those without masks are assigned to the training set, while those with masks are allocated to the validation set. This strategy enables effective evaluation of model performance under controlled conditions.

In the first training phase, which involves Noise-Imitation Learning (NIL), input images are preprocessed by converting them to grayscale and normalizing pixel values to the range [0, 1]. This standardization ensures consistency in input representation and promotes stable adversarial training.

During the second training phase, which focuses on conditional diffusion modeling, input images undergo the same grayscale conversion and normalization procedures. Additionally, affine transformations are applied for data augmentation, including random rotations, translations, and scaling. These augmentations enhance the diversity of the training data and improve the model's robustness to geometric

variations, thereby supporting more effective generalization during the reverse diffusion process.
The combination of carefully curated data, structured partitioning, and comprehensive preprocessing ensures a rigorous foundation for developing and evaluating dehazing algorithms targeted at challenging clinical scenarios.

2.2 Training Strategy

A structured two-stage training strategy is employed to optimize the proposed framework. The first stage focuses on training the Noise-Imitation Learning (NIL) component, while the second stage dedicates to training the conditional diffusion model. This phased approach ensures stable convergence and effective knowledge transfer between the two components.

During the GAN training phase within the NIL framework, the Adam optimizer is utilized with differentiated learning rates for the generator and discriminator. The generator is optimized with a learning rate of 2×10^{-4}, while the discriminator employs a lower learning rate of 1×10^{-4} to maintain training stability. Furthermore, the generator and discriminator are updated with different frequencies - the generator is updated once per iteration while the discriminator is updated twice, ensuring balanced adversarial training. A cyclical cosine annealing schedule is applied to both components, periodically restarting the learning rates according to a cosine decay pattern to help escape local optima.

For the conditional diffusion model training, the Adam optimizer is employed with an initial learning rate of 1×10^{-4}. A standard cosine annealing strategy is implemented to gradually reduce the learning rate over the training process, enabling precise convergence in the later stages of optimization. The model is trained to minimize the mean squared error between predicted and actual noise components at each diffusion step.
Both training stages utilize a batch size of 8 to maintain computational efficiency while ensuring gradient stability. The models are trained for 200 epochs each, with validation performance monitored at regular intervals to prevent overfitting. This comprehensive training strategy not only ensures stable optimization of individual components but also maximizes the synergistic effects between the GAN-based prior learning and the diffusion-based refinement process.

2.3 Results

To comprehensively evaluate the performance of the proposed method, comparative experiments were conducted on the test set containing both easy-to-image and difficult-to-image cases. The evaluation compares two configurations: (1) the baseline model utilizing only the Noise-Imitation Learning (NIL) component, and (2) the complete NIL-Diffusion framework incorporating both NIL and conditional diffusion refinement.

Table 1. Partial evaluation metrics

	CNR	gCNR	KS_A	KS_B	FID
NIL	1.737	0.788	0.092	0.675	68.33
NIL-Diff	1.790	0.799	0.098	0.685	64.62

Table 2. Downstream task evaluation metrics and final score

	Dice	ASD	Score
NIL	0.905	1.837	84.52
NIL-Diff	0.907	1.775	86.55

References

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